

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2
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ST. ANDREW'S JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY

Paper 2

Monday 29th August 2022 Total 2 hours

Materials: Question Papers

9744/2

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
 Write in dark blue or black pen on both sides of the paper.
 You may use a soft pencil for any diagram, graph or rough working.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
 Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

For Examiners' Use	
1	/10
2	/10
3	/8
4	/12
5	/10
6	/10
7	/10
8	/11
9	/8
10	/6
11	/5
Total	/100

This document consists of **29** printed pages **1** Blank page.

[Turn over]

QUESTION 1

(a) Fig. 1.1 shows an electron micrograph of a plant cell.

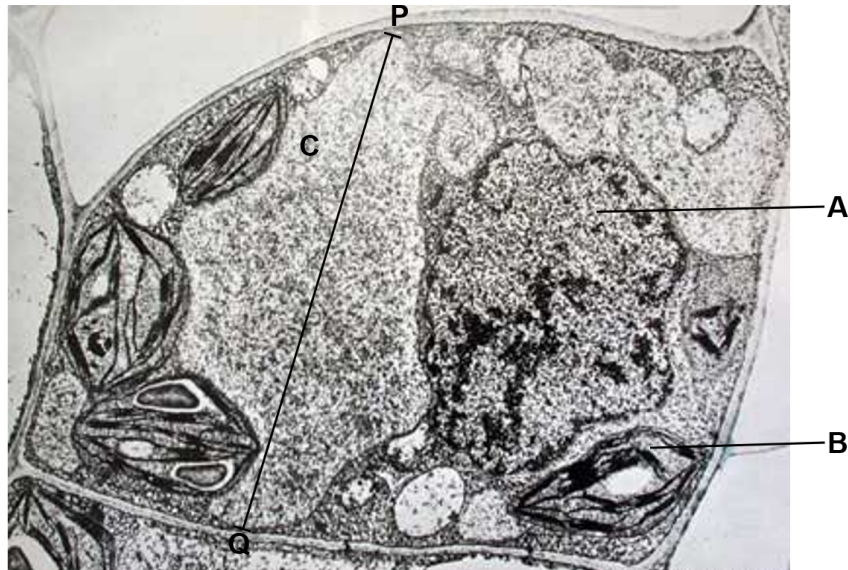


Fig. 1.1

- (i) The magnification of the photomicrograph is x560. Calculate the actual length of organelle C in μm , along the line P-Q. Show your working. [2]

Length of PQ = 6.8cm

$$\text{Actual length of organelle C} = \frac{\text{length of image}}{\text{magnification}} = \frac{6.8 \times 10\,000 \mu\text{m}}{560}$$

$$= 121.4 \mu\text{m} \text{ (1 d.p.) or } 121.43 \text{ (up to 2 d.p.)}$$

- 1 Use of correct formula and accurate measurement of length PQ
- 2 Correct number of decimal places and use of unit.

- (ii) Identify organelles **A** and **B** and compare their structural features. [4]

- 1 Organelle A: nucleus Organelle B: chloroplast [1 mark]

Similarities [max 1]:

- 2 Both of them are bound by double membrane.
- 3 Both of them contain DNA (and RNA).

Differences [max 2]:

- 4 Nucleus has linear DNA while chloroplast has circular DNA.
- 5 Nucleus does not have a third membrane system inside it/thylakoids while chloroplast has thylakoid membranes inside.
- 6 Nucleus is larger/more dense than chloroplast.
- 7 Nucleus does not contain ribosomes but chloroplast contain 70S ribosomes.
- 8 Presence of chlorophyll pigments in chloroplasts etc vs absent in nucleus

AVP: A is continuous with RER, but not B. A has pores but not B. B has electron carriers but not A. B has starch granules but not A.

- (b) Cyclins are a group of proteins that regulate the cell cycle. There are four basic types of cyclins found in humans and most other eukaryotes: G₁ cyclins, G₁/S cyclins, S cyclins, and M cyclins.

Each cyclin is associated with a particular phase, or set of phases, in the cell cycle and helps drive the events of that phase or period. For instance, M cyclin promotes the events of M phase.

The levels of the different cyclins vary considerably across the cell cycle, as shown in Fig. 1.2.

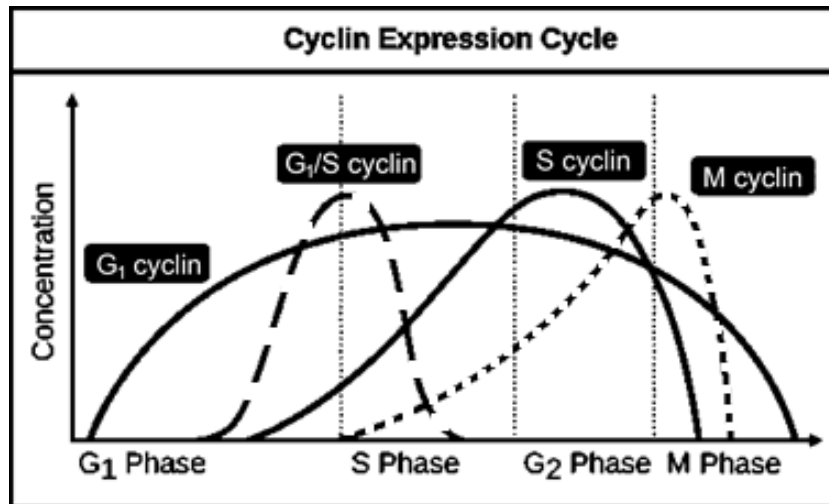


Fig. 1.2

A typical cyclin is present at low levels for most of the cycle but increases rapidly at the stage where it is needed. In order to drive the cell cycle forward, a cyclin must activate enzymes called the cyclin-dependent kinases (Cdks). In general, Cdk levels remain relatively constant across the cell cycle, but Cdk activity changes as levels of the various cyclins rise and fall.

Fig. 1.3 shows the cell cycle checkpoints and the activation periods of various Cdks. Percentage times (%T) in each phase are approximations of total cell cycle time. For instance, cells spend approximately 5% of their total cell cycle time in M phase.

Some cells in G_0 phase, such as nerve cells, never divide again. Others can divide again when called upon.

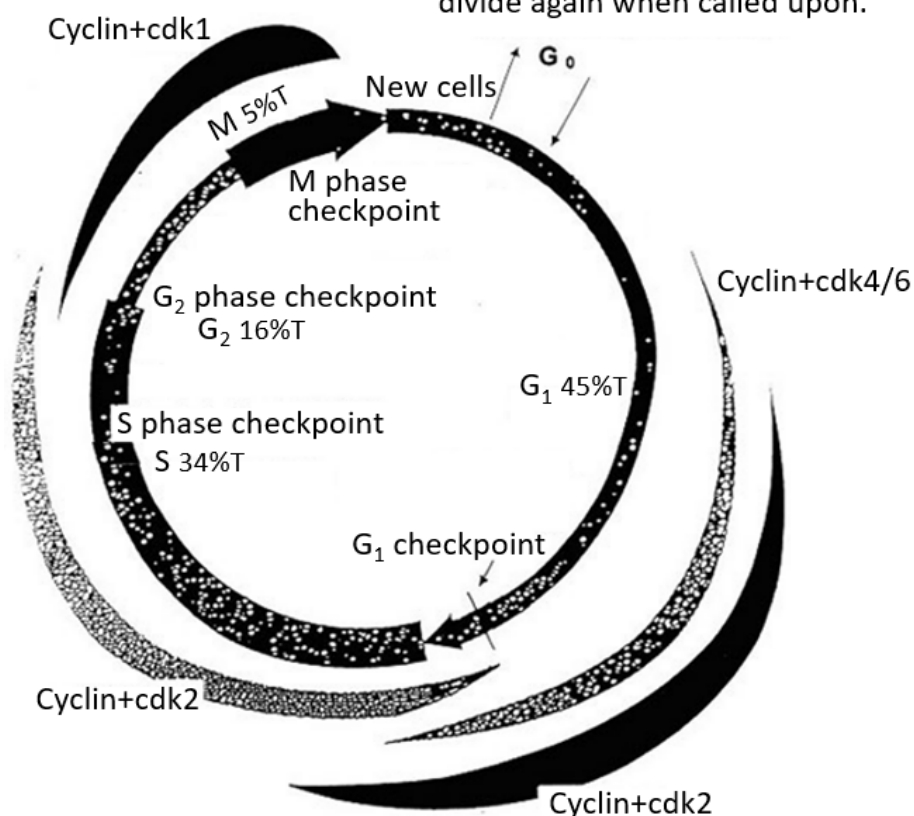


Fig. 1.3

- (i) With reference to post-translational control of gene expression, suggest how cells reduce the concentration of cyclins when they are not needed in the cell cycle.

..... [1]
 1 Ubiquitinate the cyclins to target them to proteasomes for degradation.

- (ii) With reference to Fig. 1.2 and 1.3, identify the molecules that are at abnormally low concentrations for cells to be arrested at S phase.

..... [1]
 1 M cyclin and cdk 1

Explanation: Fig. 1.2 shows that M cyclins need to increase for cells to enter G_2 , Fig. 1.3 shows cdk1 starts to increase during G_2 phase, implying that without it, cells cannot proceed to G_2 .

- (iii) It was found that cancerous cells show only 2%T at G_2 phase. Explain the implications of this.

..... [2]
 1 Cancer cells spend **lower** percentage time at G_2 phase than normal cells which implies that cancerous cells **bypass** the G_2 checkpoint
 2 Cells with DNA damage/abnormalities/replication errors are allowed to proceed to M phase / divide.
 3 allowing them to **accumulate mutations** which can lead to cancer development.

[Total: 10]

QUESTION 2

(a) Explain how the molecular structure of haemoglobin is related to its functions.

[3]

- 1 Amino acids with **charged/polar R-groups** are found on the **exterior** of haemoglobin (structure); enabling the globular protein to be soluble (Property) in aqueous medium **/cytoplasm of Red blood cell (RBC)→ can transport oxygen** to organs when RBC move through the blood stream (function)
 - 2 **One Haem group** is present in each subunit (S)→ **4 oxygen molecules** are able to be carried by each haemoglobin (P) → **increase efficient of oxygen molecules** to be carried by the haemoglobin protein (F).
 - 3 **Fe²⁺ present in each Haem grp** (S)→ allows **reversible binding of oxygen** (P) → can **release oxygen to tissues** where **oxygen level is low** (F)
 - 4 The four subunits in haemoglobin associate together **via non-covalent interactions**→ slight changes in oxygen concentration causes a **conformational change in one subunit** which is relayed to the other subunits, enabling **cooperative binding of oxygen/ easier uptake of oxygen**
- (b) Blood arriving in the lungs from the heart is oxygenated as it passes through the pulmonary capillaries. Sickle-shaped red blood cells are present in a person with sickle cell anaemia. These red blood cells have a very high quantity of abnormal haemoglobin and take up and transport less oxygen than red blood cells containing normal haemoglobin.

The cause of the differences between abnormal haemoglobin and normal haemoglobin is a mutation in the gene that codes for one of the two types of polypeptide found in a haemoglobin molecule. This mutation leads to a change in the mRNA produced during transcription, causing a change in the primary structure of the polypeptide formed.

Fig. 2.1 shows some of the changes that occur as a result of this gene mutation.

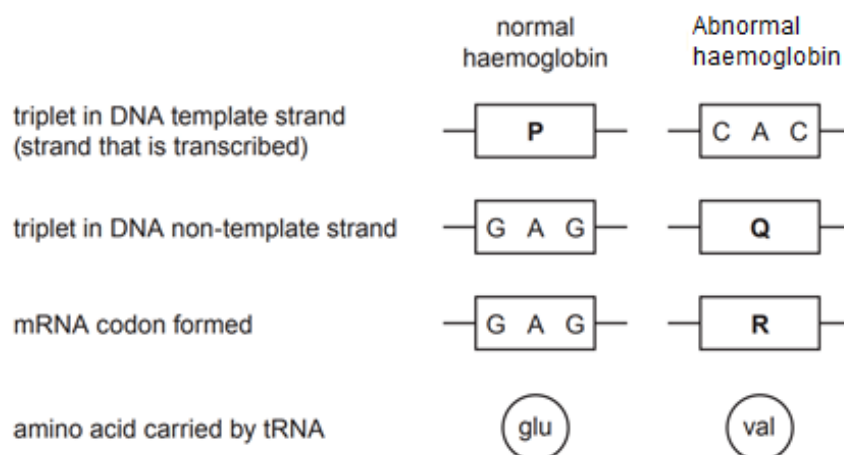


Fig. 2.1

- (i) With reference to Fig. 2.1, state:
- the base sequence of DNA triplet P

.....
P - C T C ;

- the base sequence of DNA triplet Q

.....
Q - G T G ;

- the base sequence of mRNA codon R

.....[3]
R - G U G ;

- (ii) Name the polypeptide that is altered in the abnormal haemoglobin molecule.

.....[1]
 β -globin chain / beta globin;

(c) Fig. 2.2 shows the oxygen dissociation curve for adult haemoglobin in a person who does not have sickle cell anaemia.

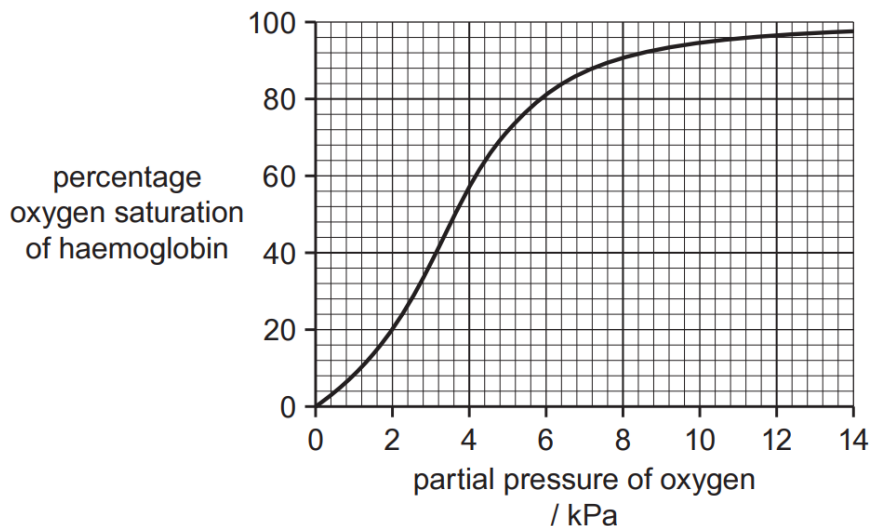


Fig. 2.2

Compared to Fig. 2.2, the oxygen dissociation curve for adult haemoglobin in a person with sickle cell anaemia is shifted to the right. The uptake of oxygen by haemoglobin in the lungs and the release of oxygen by haemoglobin in respiring tissues is different in a person with sickle cell anaemia compared with a person who does not have the disease.

With reference to Fig. 2.2, state **and** explain these differences.

uptake of oxygen

.....

release of oxygen

.....

Explanation for these differences

.....[3]

any three from:

in terms of sickle cell:

uptake:

1 lower uptake of oxygen / lower saturation of haemoglobin at same partial pressures/e.g. requires higher partial pressure of oxygen to reach same level of saturation

Release:

2 oxygen is more easily released (at same partial pressures);/ Oxygen release at higher partial pressure of oxygen

3 Explanation : lower affinity for oxygen / lower carrying ability of haemoglobin S for oxygen / more difficult for oxygen to bind to haemoglobin S / AW + due to any of the following

- ref. to structure of abnormal haemoglobin, e.g. form fibres at low oxygen concentration
- reduced allosteric effect of haemoglobin molecule for uptake
- ref. to allosteric release of oxygen in respiring tissues

[Total: 10]

QUESTION 3

HER2-positive breast cancer is a breast cancer that tests positive for a protein called human epidermal growth factor receptor 2. A member of the HER family of tyrosine kinase receptors (HER1–4), HER2 is an essential breast cancer oncogene.

Fig. 3.1 shows the Southern blot (DNA analysis) and Western blot (protein analysis) results of HER2 gene expression for a healthy individual and a breast cancer patient, and Fig. 3.2 shows the HER2 signaling pathway.

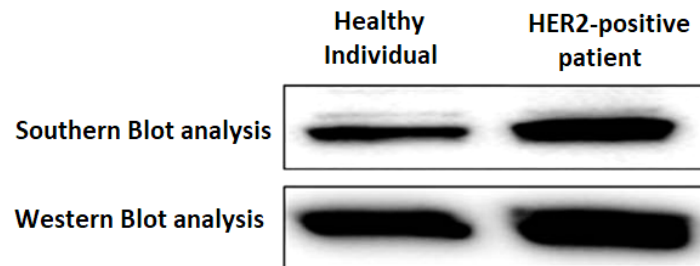


Fig. 3.1

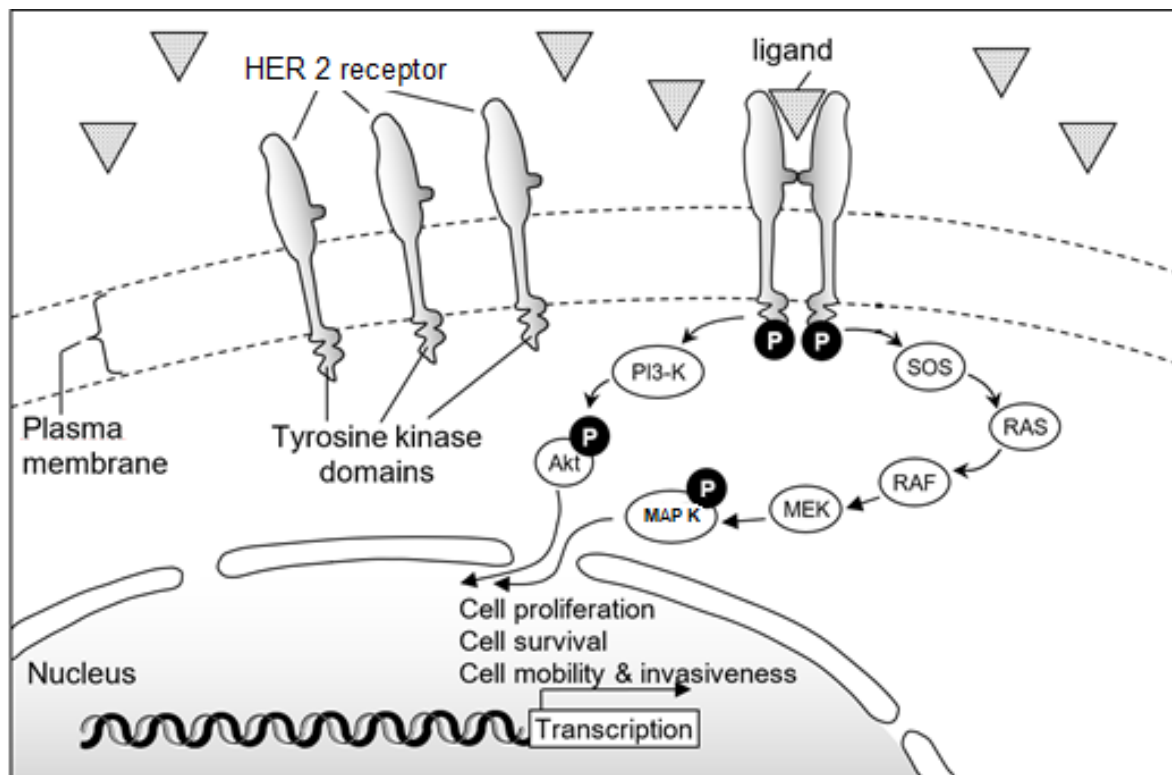


Fig. 3.2

- (a) Research shows that 1 in 5 cancer patients shows an over-expression of HER2 gene. With reference to Fig. 3.1, suggest how the HER2 gene is converted to an oncogene in cancer patients.

[1]

(Thicker band seen for the Southern Blot analysis)

1 ref. Amplification of / Increase in number of copies of HER2 gene;

R! All other scenarios

- (b) Distinguish between how the HER2 receptor and a G-protein linked receptor function as transmembrane receptors upon binding of their respective ligands.

[2]

- 1 Binding of ligand leads to dimerization of HER2 receptors, but no dimerization for GPLR;
- 2 Activation of HER2 receptors results in auto-cross phosphorylation of the tyrosine kinases domains, but activation of GPLR results in activation of G-protein.

- (c) Patients with HER2-positive breast cancer have responded well to treatment when the drug Herceptin is used in combination with other chemotherapy drugs. Herceptin binds to the extracellular domain of the receptor.

With reference to Fig. 3.2, suggest and explain how Herceptin may act to prevent further progression of breast cancer.

[3]

- 1 ref. prevents ligand-receptor interaction via
 - Drug is complementary in shape to / binds at / competes with ligand for binding at ligand-binding site of HER2 receptor
 - / Drug binds at a site other than the ligand-binding site, and changes 3D conformation of ligand binding site;
- 2 ref. preventing downstream signal transduction such as activation of SOS, RAS, RAF, MEK, MAP K etc, and cellular response
- 3 ref. at least one of cellular response - cancer cells die / stop dividing / no loss of anchorage dependence / unable to undergo metastasis;

- (d) Even with the effective treatment of Herceptin and chemotherapy, a proportion of cancer patients who were in remission, still experienced recurrence of cancer some years later.

Suggest how the cancer cells in these cases overcome the effects of Herceptin.

[2]

- 1 ref. gene (substitution) mutation of HER2 gene resulting in changes of 3D conformation of ligand-binding site / allosteric site;
- 2 Unable to bind to Herceptin;

OR

- 1 Gain-in-function mutation of downstream relay proteins to become constitutively active / no longer dependent on activation of HER2 receptor;
- 2 constitutive signal transduction → expression of genes resulting in increased cell proliferation, independent of ligand binding.

OR

- 1 **Gene amplification** / over-expression of genes coding for ;
- 2 enzyme/protein that degrades/hydrolyses Herceptin (prevents it from binding to HER2 receptor;

AVP

- 1 Gene amplification of the HER2 gene / chromosomal translocation of gene near enhancer/more active promoter → increase expression of HER2 gene,
- 2 Increase in HER2 receptors embedded in membrane => overcome inhibitory effect of Herceptin (present in lower concentrations)

[Total: 8]

QUESTION 4

- (a) Fig. 4.1 shows the small ribosomal subunit rRNA loops containing double-stranded sections, H27, H30, H31 and H34, from bacteria and the mitochondria of mammals and plants. Grey shading highlights the locations of H31 and H34.

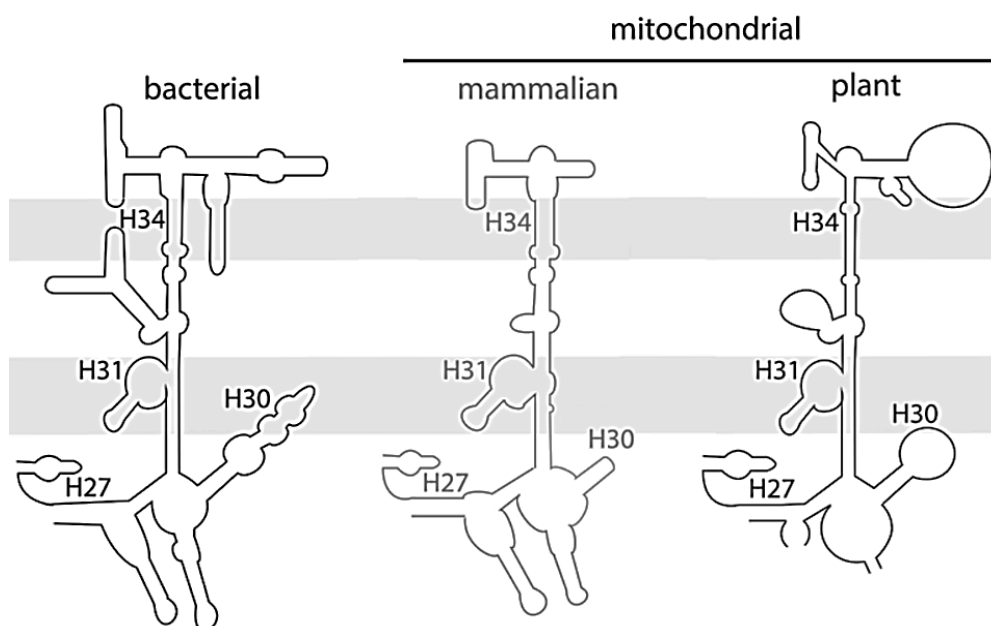


Fig. 4.1

- (i) With reference to Fig. 4.1, **compare** the structures of the bacterial and mammalian mitochondrial small ribosomal subunit rRNA. [2]

Similarities

- 1 Both are single-stranded rRNA.
 - 2 Both have similar loop structures at H27, H31 and H34 (need at least 2)
- AVP: Both contain H27, H30, H31 and H34 double-stranded helices.

Differences

- 3 The H30/H34 in bacterial mitochondrial rRNA has a longer/more extensive loop that is absent from the H30/H34 in mammalian rRNA.
- 4 The bacterial rRNA is longer than the mammalian mitochondrial rRNA.

- (ii) Describe the roles of rRNA in the process of protein synthesis. [2]

- 1 rRNA is **assembled with proteins** to form ribosomal subunits, which assemble to form ribosomes.
- 2 hold tRNA and mRNA in close proximity, to position new amino acid to growing polypeptide chain
- 3 it has peptidyl transferase activity, to catalyse formation of peptide bond during translation

- (iii) Mammals and plants belong to the domain Eukarya while bacteria are grouped into their own domain, Bacteria. With reference to Fig. 4.1, comment on the suitability of using the small ribosomal rRNA structure to study the phylogenetic relationships between bacteria, mammals and plants.

..... [3]

- 1 Not suitable.
- 2 Mammals and plants share a more recent common ancestor than they do with bacteria / mammals and plants are more closely related to each other than either one is to bacteria.
- 3 However, the **loop structures** of the small ribosomal rRNA of mammals are more similar to those of bacteria than to plants.

AVP: determining the extent of similarity/differences in the loop structures can be subjective

AVP: as the rRNA from mammalian and plant cells were taken from their mitochondria, which are **prokaryotic** in origin, it is not suitable to compare their structure with bacteria rRNA to study phylogenetic relationship between Bacteria and Eukarya. Instead, eukaryotic rRNA from mammalian and plant cell cytoplasm should be used.

- (b) Fig. 4.2A shows an electron micrograph of DNA replication occurring in organism X. Fig. 4.2B shows a representative diagram of the same process. The dotted line represents the daughter strands. Ori R represents the origin of replication.

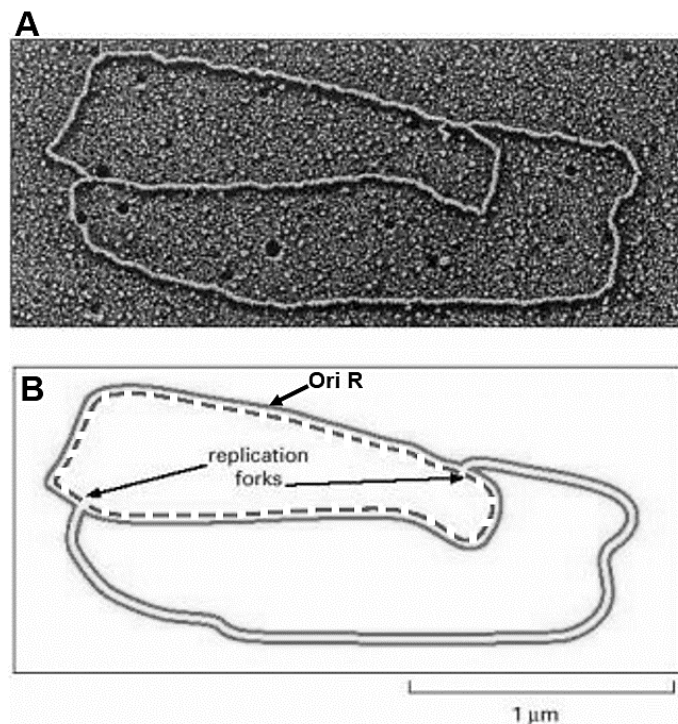


Fig. 4.2

- (i) State whether organism X is a prokaryote or a eukaryote. Explain your answer.

..... [2]

- 1 Prokaryote
 - 2 It has circular DNA.
- AVP one origin of replication

Fig. 4.3 shows a diagrammatic representation of the DNA replication process. The arrow in the diagram on the left shows the direction of the synthesis of one of the daughter strands.

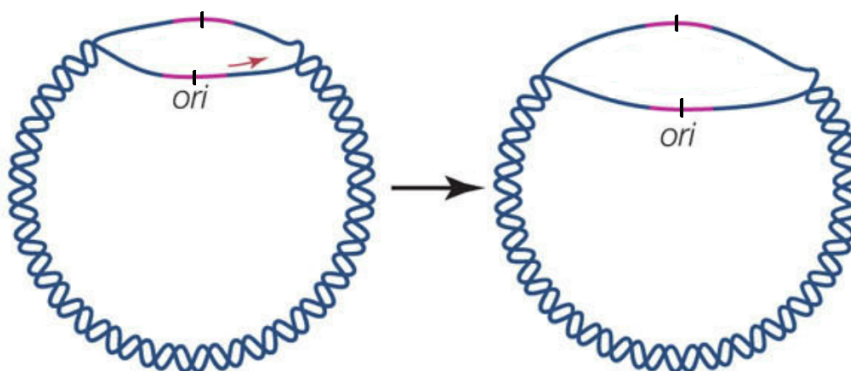
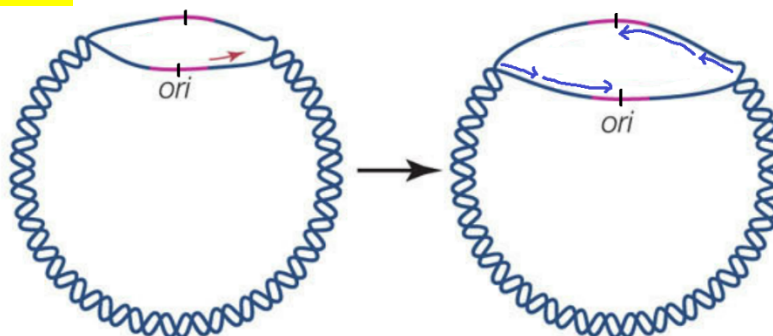


Fig. 4.3

- (ii) Draw arrows in Fig. 4.3, on the DNA to the right, to indicate how the Okazaki fragments are synthesised. [1]

ANS:



- (iii) Describe how the process of DNA replication differs between prokaryotes and eukaryotes. [2]

Feature	Prokaryotes	Eukaryotes
Number of origins of replication	DNA replication begins at a single origin of replication .	DNA replication begins at multiple origins of replication .
Location	Occurs in the nucleoid region.	Occurs in the nucleus .
Replication complete or incomplete	The whole chromosome is replicated completely.	The 3' ends of parental strands are not copied onto the 5' end of daughter strands.
Speed	Relatively fast (due to smaller DNA)	Relatively slower (due to larger DNA molecules)

[Total: 12]

QUESTION 5

Gibberellins (GA) are plant hormones that regulate developmental processes such as stem elongation and flowering. Fig. 5.1 shows the effect of different levels of GA on stem elongation. Plant 1 lacks GA and has an internode length of "0" ("dwarf" plant). Plant 2 has a moderate amount of GA and an average internode length ("normal" plant). Plant 3 has a large amount of GA and so has a much longer internode length ("giant" plant).

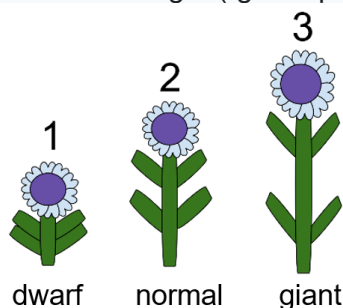


Fig. 5.1

Gene **K** encodes an enzyme responsible for one of the first steps of GA biosynthesis in *Arabidopsis* plants. The recessive allele of this gene results in GA-deficient *Arabidopsis* dwarfs. The dominant allele of Gene **T** codes for an activator which triggers the over-expression of Gene **K**, resulting in *Arabidopsis* giants.

When pure-breeding normal plants are crossed with pure-breeding dwarf plants, they produced F_1 which are all giants. Selfing of the F_1 generation produced the following F_2 generation:

443 giants
159 normal
197 dwarfs

(a) Draw a genetic diagram to show the results of the F_2 generation from the selfing of the F_1 generation.

..... [3]

F_1 phenotype
 F_1 genotype
Gametes

Arabidopsis Giant
KkTt

x

Arabidopsis Giant
KkTt

x

(KT) (Kt) (kT) (kt)

(KT) (Kt) (kT) (kt)

F_2 Punnett square

	(KT)	(Kt)	(kT)	(kt)
(KT)	KKTT giant	KKTt giant	KkTT giant	KkTt giant
(Kt)	KKTt giant	KKtt normal	KkTt giant	Kktt normal
(kT)	KkTT giant	KkTt giant	kkTT dwarf	kkTt dwarf
(kt)	KkTt giant	Kktt normal	kkTt dwarf	kktt dwarf

F_2 phenotypic ratio 9 giants : 3 normal : 4 dwarfs

Mark Scheme

- 1 F_1 genotype match with phenotype
- 2 Gametes (circled)
- 3 F_2 genotypes match with phenotypes + correct ratio

- (b) (i) The expected phenotypic ratio of the F₂ generation is 9 giants : 3 normal : 4 dwarfs.

Calculate the value of χ^2 applicable to these data.

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

ν = degrees of freedom

E = expected 'value'

c = number of classes

[2]

- 1 Calculation of expected numbers:

Classes	Observed number (O)	Expected ratio	Expected number (E)
Giant	444	9	449
Normal	159	3	150
Dwarf	197	4	200

- 2 Calculation of χ^2 value:

$$\chi^2 = \frac{(443-449)^2}{449} + \frac{(159-150)^2}{150} + \frac{(197-200)^2}{200}$$

$$= 0.67 \text{ (2 d.p.)}$$

[Accept: 0.70 for students who gave expected numbers to 1 d.p.]

- (ii) Use the calculated value of χ^2 and the table of probabilities provided to find the **probability** of the observed results differing by chance from the expected numbers.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

[2]

- 1 Degrees of freedom = 3-1 = 2

- 2 $p > 0.10$

- (iii) State the conclusions that may be drawn from the probability found in (ii).

[3]

- p value is larger than 0.05, there is no significant difference between observed and expected results. Any difference is due to chance.
- There is independent assortment of Genes K and T.
- kk is epistatic over the T/t locus, resulting in dwarf plants due to the lack of the enzyme involved in GA synthesis.
- In the presence of a dominant K allele, the dominant T allele results in over-expression of the K allele, resulting in giant plants.

[Total: 10]

QUESTION 6

Bacteria can undergo genetic recombination, a process by which genetic information from one bacterium is transferred to, and then recombined with that of another bacterium.

Experiments have shown that two strains of bacteria can undergo conjugation when grown in a common medium. However, when they are separated by a filter in the Davis U-tube, as shown in Fig. 6.1, conjugation cannot occur.

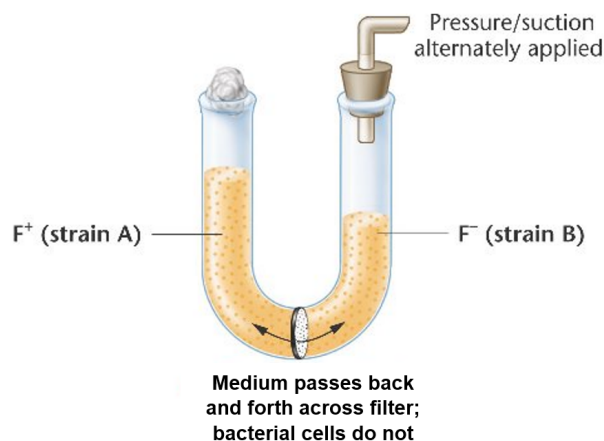


Fig. 6.1

(a) With reference to Fig. 6.1, explain why conjugation cannot occur when the two strains of bacteria are placed in the Davis U-tube.

- [2]
- 1 Ref to presence of filter therefore **no cell to cell contact** between strain A and strain B;
 - 2 **No attachment via sex pilus / no formation of conjugation / mating bridge** therefore no transfer of plasmid from F⁺ donor cell to F⁻ recipient cell;

In another experiment investigating possible recombination in the bacterium *Salmonella typhimurium*, when researchers mixed *Salmonella* strains X and Y in a common medium, they recovered recombinants.

In a separate experiment using the Davis U-tube, the strains X and Y were separated by a filter, thus preventing cell contact but allowing growth to occur in a common medium. Surprisingly, when samples were removed from both sides of the filter, recombinants were recovered only from the side of the tube containing strain X bacteria. Researchers postulated that a filterable agent was released by the strain Y cells and responsible for transferring the new genetic information.

Three subsequent observations were useful in identifying the filterable agent:

1. The filterable agent was released by the strain Y cells only when they were grown in association with strain X cells.
2. The addition of DNase, which enzymatically digests naked DNA, did not render the filterable agent ineffective.
3. The filterable agent could not pass across the filter of the Davis U-tube when the pore size was reduced below the size of 20nm.

(b) (i) Suggest the source of the new genetic information which strain X cells received.[1]

- **Strain Y; [Reject: virus]**

(ii) State the process of genetic recombination that has occurred. [1]

- **Transduction;**

(iii) Describe how the filterable agent could have transferred the new genetic information to strain X. **[3]**

1 (Generalised transduction) Phage enzymes degrade bacterial host DNA and parts of bacterial DNA repackaged into new phages ;

OR

2 (Specialised transduction) Phage DNA integrate to form prophage, but later excise part of bacteria DNA out before repackaging into new phages;

3 Progeny phages released from the host carrying Strain Y DNA cross filter attaches to Strain X bacterial cell wall (receptor) / to infect other recipient strain X bacteria;

**4 Ref. to homologous recombination/site specific integration
NO ECF**

Fig. 6.2 represents the reproductive cycle that a T4 phage can carry out.

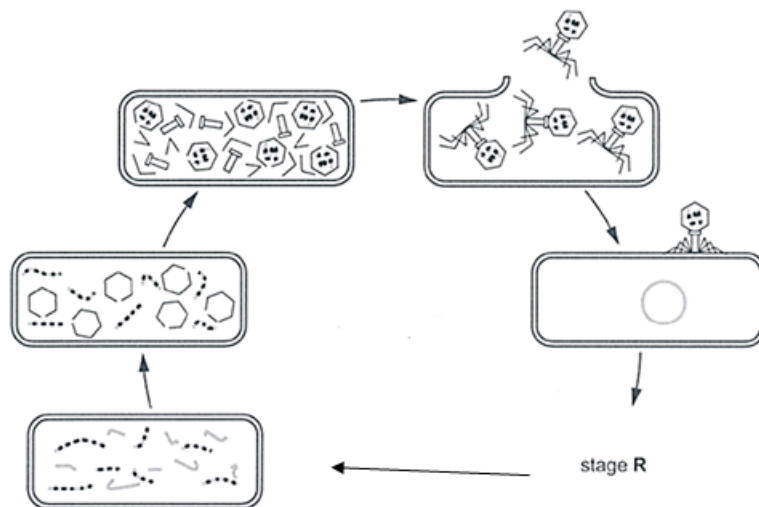


Fig. 6.2

(c) Describe what happens in stage R of the reproductive cycle shown in Fig. 6.2.[3]

1. Tail sheath contracts and injects the phage DNA genome through the cell wall and plasma membrane
2. Ref. host DNA/chromosome broken down to provide nucleotides
3. ref. to synthesis of viral protein e.g. capsid protein using host ribosomes
4. Ref. viral genome using host DNA polymerase;

[Total: 10]

QUESTION 7

Corals grow in shallow sea water. Corals consist of colonies of small animals called polyps. These polyps have photosynthetic protists called algae within their cells, which is advantageous both to the coral polyps and to the algae.

The algae that live within the cells of coral polyps can also live independently as free-living algae.

- (a) The rate of photosynthesis of algae that live within the cells of coral polyps is higher than that of free-living algae.

Suggest and explain why the rate of photosynthesis in algae that live within the cells of coral polyps is higher than that of free-living algae.

.....[2]

- 1 The carbon dioxide concentration is higher / increased within the cells of the coral due to **respiration** from coral / polyp ;
 - 2 hence there is increased CO₂ for Calvin cycle / light independent reactions in algae ;
- AVP** further detail of effect of CO₂ concentration on binding of CO₂ to rubisco ;

The algae that live within the cells of coral polyps have five different chloroplast pigments.

Table 7.1 shows the light wavelengths at which each algal chloroplast pigment shows its two largest peaks of light absorption.

Table 7.1

chloroplast pigment	peak 1 wavelength / nm	peak 2 wavelength / nm
chlorophyll a	430	662
peridinin	456	485
chlorophyll c ₂	450	396
dinoxanthin	442	471
β-carotene	454	480

Corals can be kept in glass tanks that are usually lit by lamps radiating mainly violet and blue light, with wavelengths in the range of 400 nm to 490 nm.

With reference to Table 7.1, suggest why lamps radiating mainly violet and blue light are expected to increase the growth of coral polyps more than lamps radiating light of all wavelengths.

..... [3]

- 1 pigments absorb violet-blue light with wavelength 400–490 nm well / best / most / at 8 (out of 10) peaks ;
- 2 rate of photosynthesis of algae increases with more light absorbed ;
- 3 ref. to transfer of, organic nutrients, to polyps from algae and hence more coral growth (increases) with more photosynthesis ;

- (c) ATP is used or produced at different stages in the oxidation of glucose in aerobic conditions. Complete Table 7.2 to show whether ATP is used or produced at each stage of respiration. Write either YES or NO in each box. [2]

Table 7.2

stage of respiration	ATP used	ATP produced
glycolysis		
link reaction		
Krebs cycle		
oxidative phosphorylation		

stage of respiration	ATP used	ATP produced
glycolysis	yes	yes
link reaction	no	no
Krebs cycle	no	yes
oxidative phosphorylation	no	yes

;;

4 correct = 2 marks, 2 or 3 rows correct = 1 mark

If ticks and crosses used need all 4 correct for maximum 1 mark

- (d) An experiment was carried out to investigate the effect of epicatechin on mitochondrial respiration in mice. Epicatechin is a naturally occurring compound in cocoa beans and so is present in chocolate. Two groups of mice, group A and group B, were used in this experiment.

- Group A was given water containing epicatechin, twice a day for 15 days.
- Group B was given water without epicatechin, twice a day for 15 days.

After 15 days, the structure of mitochondria from striated muscle cells in both groups of mice was examined. The surface area of the inner membrane of the mitochondria was divided by the surface area of the outer membrane to obtain a ratio for each mouse. Table 7.3 shows the mean ratios for the two groups of mice.

Table 7.3

group	mean ratio
A	2.0 : 1
B	1.7 : 1

The mice in group A were observed to be able to exercise longer than the mice in group B.

With reference to Table 7.3, explain why the mice in group A were able to exercise for longer than the mice in group B.

.....[3]

- 1 Group A has **higher** mean ratio of inner membrane to outer membrane 2.0 : 1 as compared to group B with 1.7 :1 ;
- 2 This suggest that inner membrane / cristae is more highly folded than B, therefore more, ETCs / cytochromes / ATP synthetase are embedded.
- 3 resulting in higher rate of oxidative phosphorylation / chemiosmosis, and more ATP produced per unit time, hence muscles can contract for, longer / more time / without getting tired;

[Total: 10]

QUESTION 8

- (a) A geneticist studied the following pedigree diagram in Fig. 8.1 of a family with several members diagnosed with a genetic condition which originated from a mutation in a particular gene.

Given that the disease is autosomal dominant, Southern Blotting was also performed using genomic DNA isolated from each family member. The DNA samples were first subjected to restriction digestion by *EcoRI*, before hybridising to probes derived from the said gene.

The results for all family members, except those for the foetus, are shown below in Fig. 8.1. The sizes of the DNA fragments obtained after *EcoRI* digest are as indicated.

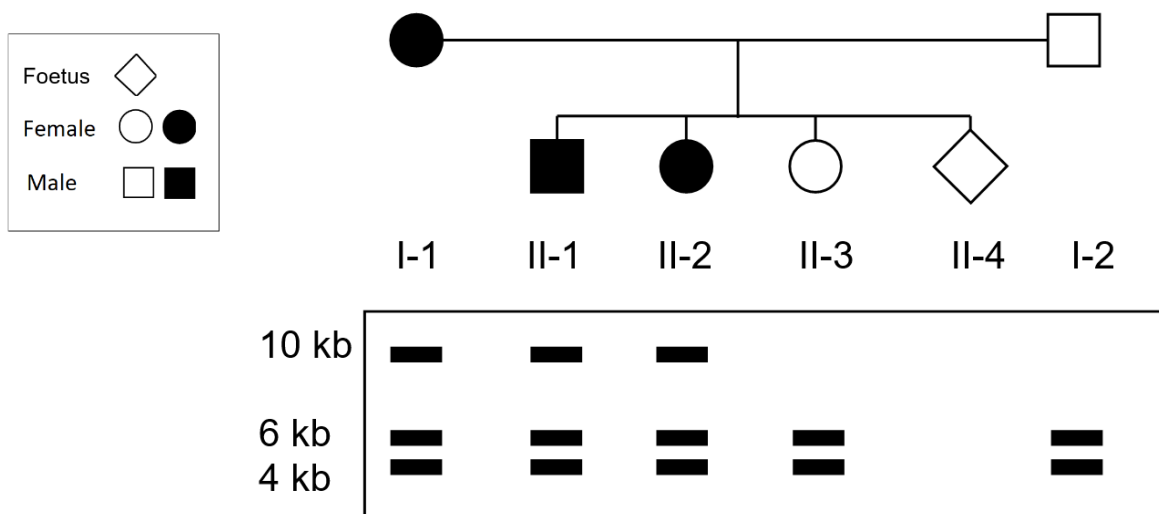


Fig. 8.1

- (i) With reference to the Southern Blot results of individual **I-1** in Fig. 8.1, use arrows to indicate the relative positions of the *EcoRI* restriction sites on the region of the genomic DNA recognised by the probe.

Label ALL the restriction sites as “R” and indicate the length of bases in kb.

.....[2]

allele 1 _____

allele 2 _____

Answer: Award mark for showing restriction sites at both ends of the alleles. 1 mark each for allele 1 and 2



- (ii) Calculate the probability of the foetus being a boy and inheriting the disease. Explain your answer.

.....[2]

- 1 ref. mother is heterozygous and father is homozygous recessive;
- 2 Probability of foetus receiving dominant allele is $\frac{1}{2}$ and probability of getting a son is also $\frac{1}{2}$;
- 3 $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ or 25%;

Accept: Punnett Square with defined alleles as explanation for 1 & 2;

- (b) Fig 8.2 shows the possible restriction sites of a gene and the results of four individuals after subjecting their genomic DNA to Polymerase Chain Reaction (PCR) and restriction enzyme digest. The number below each band indicates the molecular sizes of the individual fragments.

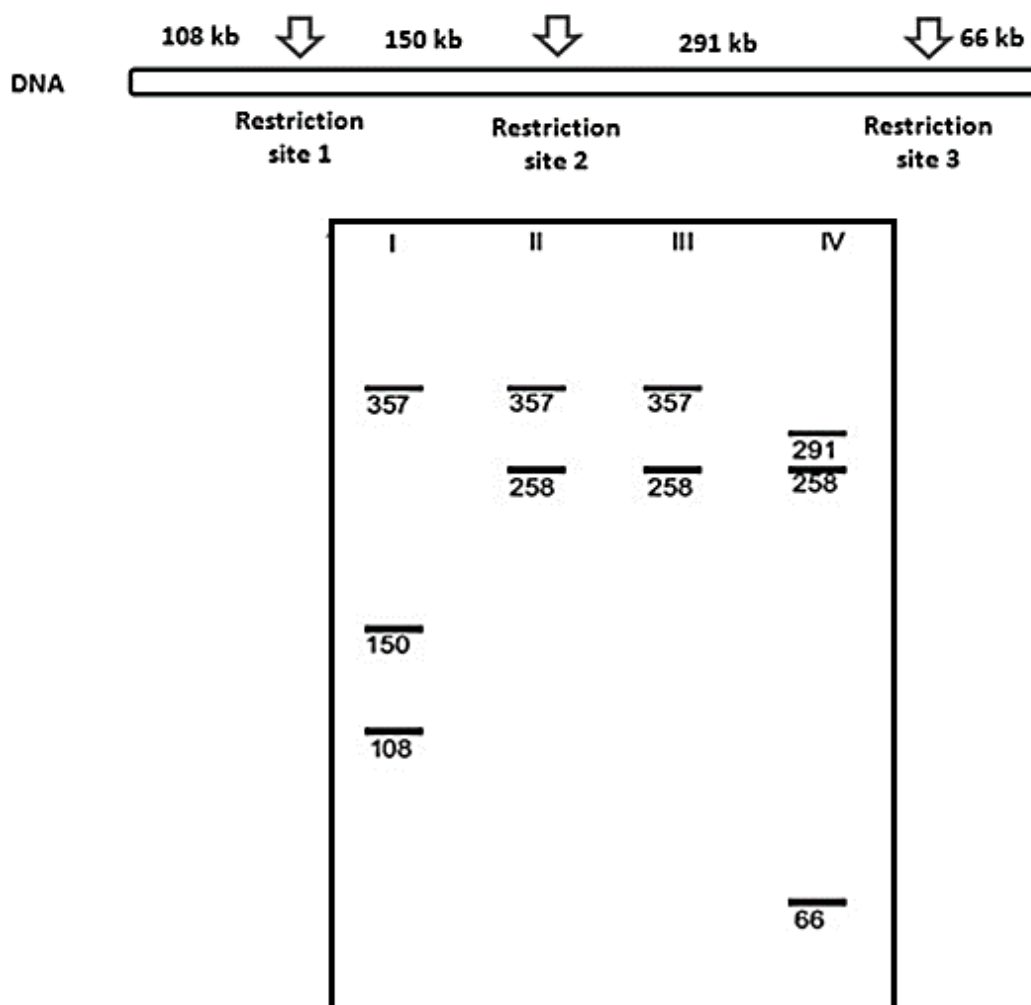


Fig 8.2

- (i) Suggest the significance of carrying out PCR for the genomic DNA samples before subjecting the samples to restriction enzyme digest.

.....[1]

- 1 To avoid getting a smear/multiple bands of DNA fragments of different molecular sizes by amplifying/isolating only the desired/target gene for analysis;

- (ii) With reference to Fig. 8.2, identify the total number of alleles presented by this gene in the four individuals tested. Explain your answer.

.....[4]

- 1 3 different alleles;
- 2 Allele 1: Restriction sites 1 and 2 resulting in 357 kb, 150 kb and 108 kb fragments in individual I;
- 3 Allele 2: Only restriction site 2 resulting in 357 kb and 259 kb fragments shown in individual II and III;
- 4 Allele 3: Restriction sites 2 and 3 resulting in 357 kb, 150 kb and 108 kb fragments in individuals IV;

- (iii) Suggest how a mutation can result in the different alleles for this gene.

.....[1]

- 1 Single base substitution that results in the loss of a restriction site;

Reject: ref. missense or nonsense mutations.

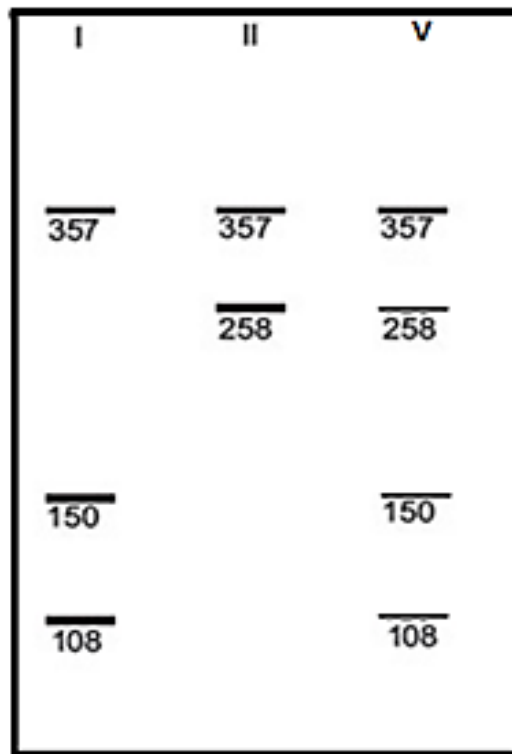
Reject: All other types of gene mutation as insertion or deletion of a single base, will change the length of the restriction fragment obtained.

- (iv) Individuals I and II had a child V. Draw his banding pattern in the diagram below after subjecting his genomic DNA to the same PCR and restriction digest treatment.

[1]

I	II	V
<u>357</u>	<u>357</u>	
	<u>258</u>	
<u>150</u>		
<u>108</u>		

Ans:



Award mark only if band for 258kb, 150kb and 108 kb are thinner compared to the homozygous parents.

[Total: 11]

QUESTION 9

The genus *Heliconius* contains more than 40 species of brightly patterned butterflies. Researchers have investigated in the laboratory how one species, *Heliconius heurippa*, could have developed as a separate species. The phenotype of *H. heurippa* is intermediate between that of two other species, *H. cydno* and *H. melpomene*.

Laboratory breeding experiments showed that:

- matings between *H. cydno* and *H. melpomene* (parent species) produce fertile hybrid offspring
- controlled matings of the hybrids produce individuals identical in appearance to *H. heurippa* within three generations
- hybrid butterflies prefer to mate with each other, rather than with individuals of either of the parent species.

(a) The researchers concluded that the *H. heurippa* species could contain DNA from the two parent species as a result of hybridisation.

(i) Suggest, with reasons, one prediction that can be made about the chromosome numbers of *H. cydno* and *H. melpomene*.

..... [2]

- 1 [Compulsory point]: Chromosome numbers are the same.
- 2 Hybrids are fertile.
- 3 Homologous chromosomes can pair up / bivalents can form (in meiosis) to produce viable gametes.

(ii) The researchers thought that, because the hybrid butterflies preferred to mate with each other, this could make speciation more likely to occur.

Give reasons why the researchers thought that this made speciation more likely.

..... [4]

- 1 There is behavioral isolation (between hybrids and parent species.)
AVP: Sexual selection
- 2 This **reduces the gene flow** between hybrids and parent species.
- 3 Genetic differences can accumulate via mutation, genetic drift and natural selection
[any two]
- 4 Over time, sympatric speciation can occur.

(b) *Heliconius* butterflies taste unpleasant to predators such as birds. The bright colours on the wings act as warnings so that birds avoid eating them.

Individual birds learn which patterns to avoid. If one *Heliconius* species is abundant, or if it has a pattern shared with another similar species, predators learn to avoid this pattern faster. Therefore, this pattern provides a selective advantage.

Fig. 9.1 shows the distribution of several *Heliconius* species in South America.

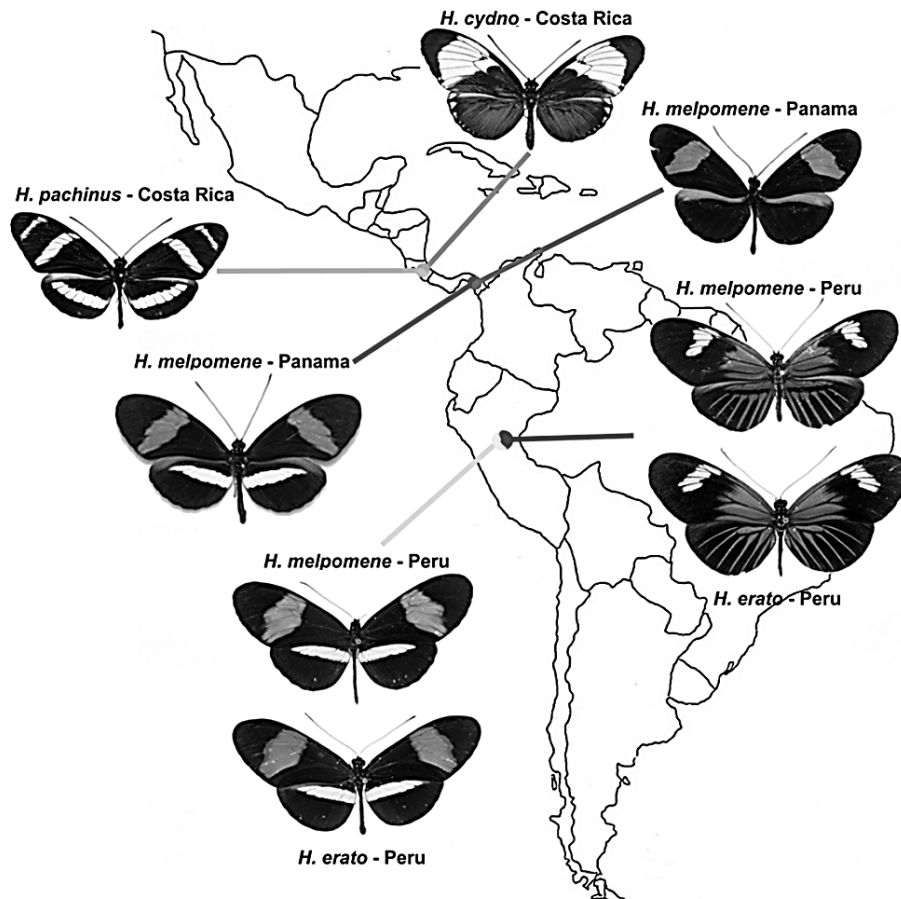


Fig. 9.1

With reference to Fig. 9.1 and your knowledge of anatomical homology and biogeography, comment on the evolutionary relationships between *H. Melpomene*, *H. cydno*, *H. pachinus* and *H. erato*.

- [2]
- 1 *H. melpomene* is very closely related to *H. erato* as they have very similar wing patterns and are found in the same locations in Panama and Peru. Accept same ecological niche / face same selection pressures;
 - 2 *H. cydno* and *H. pachinus* are more closely related to each other than they are to *H. Melpomene* as they are found in the same location in Costa Rica, but are further away from *H. Melpomene*
OR
H. pachinus is more closely related to *H. melpomene* than it is to *H. cydno* as its wing pattern more resembles that of *H. Melpomene*.

[Total: 8]

QUESTION 10

Fig. 10.1 shows some of the general changes in the immune system during a viral infection.

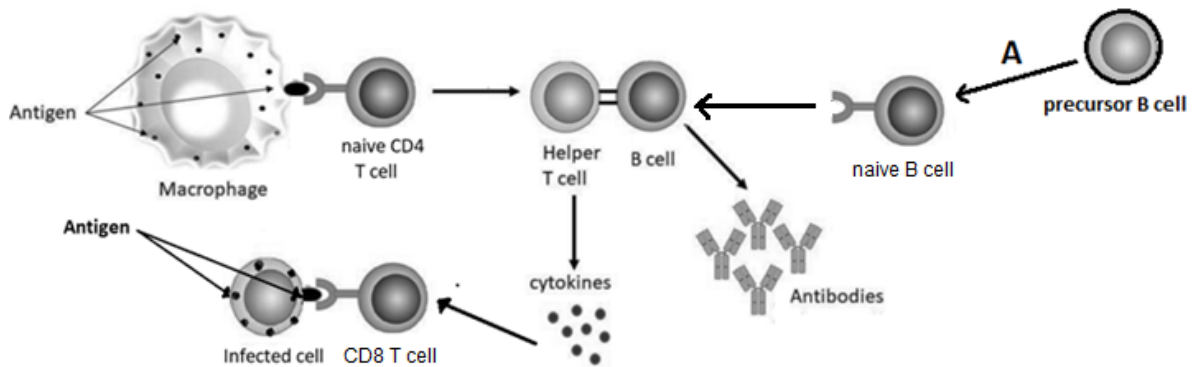


Fig. 10.1

- (a) Outline the **genetic mechanism** that occurs in **stage A** and explain its significance in the adaptive immune response.

-[4]
- 1 ref. **somatic recombination** of **V, D and J gene segments** of the variable region of the immunoglobulin genes / of **V, D and J gene segments** at **heavy chain locus** and **V and J segments** at a **light chain locus**
 - 2 giving rise to different light and heavy chain variable regions, which generates a **wide diversity of B cells** such as B1-B3; **R: antibodies**
 - 3 Each B cell has a **distinct/ specific B cell receptor** which recognises a single **specific/complementary antigen/epitope** from **different pathogens**;
 - 4 Ref. production of **specific** antibodies to neutralise/opsonise the pathogen.

- (b) A HIV-infected person infected saw his CD4 T cell count decrease within the first few weeks of his infection and then start to increase again.

With reference to Fig 10.1 and your own knowledge on adaptive immunity, suggest an explanation for the changes in his CD4 T cell count.

-[2]
- 1 [decrease in CD4 T cells] ref. infected CD4 T cells present antigens on their cell surface (via MHC Class I) and are removed/destroyed by **cytotoxic T cells** / bound by **antibodies** and destroyed by macrophages/NK cells;
 - 2 [increase in CD4 T cells] ref. HIV in latent phase, healthy CD4 T cells are being regenerated (from haematopoietic stem cells) / antigen not displayed on cell surface of infected T cells, thus T cells not lysed or killed;

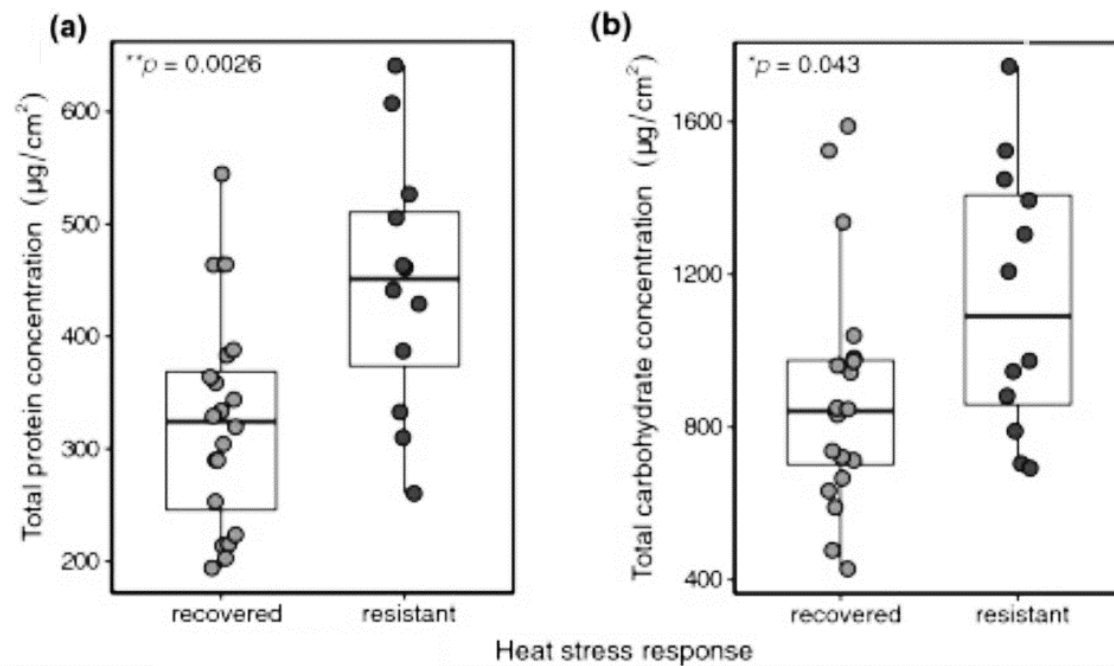
[Total: 6]

QUESTION 11

Rising seawater temperatures are contributing to coral bleaching, with mass coral bleaching events projected to increase in both frequency and severity.

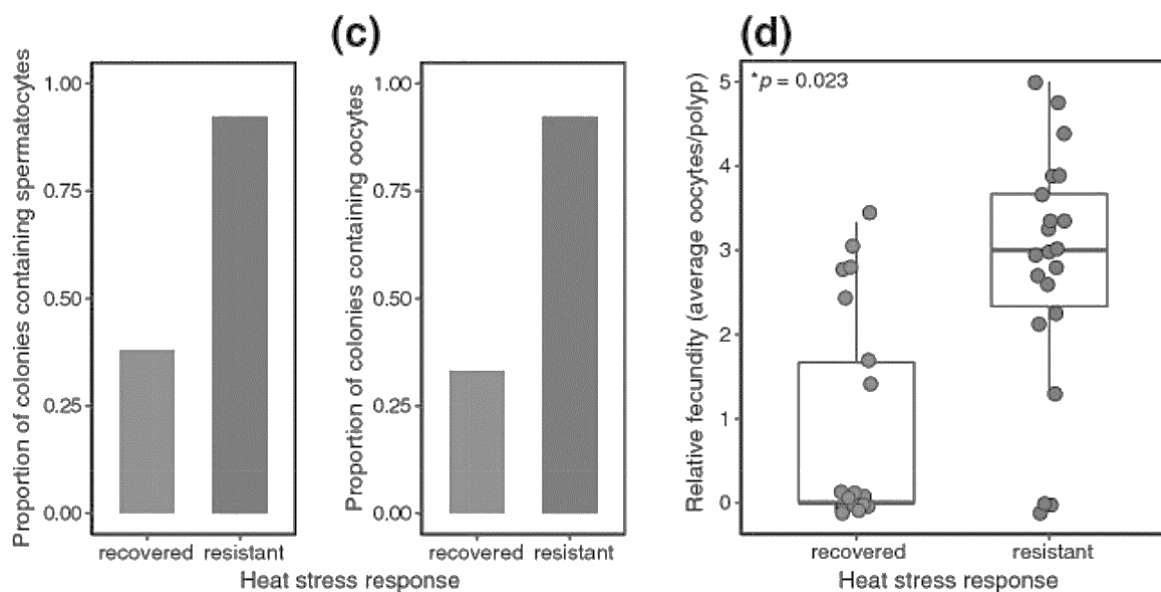
Fig 11.1 shows the impact of thermal bleaching stress on stored protein and carbohydrate reserves and reproductive output of two categories of corals five months after the bleaching event – corals that were resistant to bleaching events and corals that were bleached but later recovered.

For both types of corals, zooxanthellae was able to recolonise them and both types of corals also looked visually healthy.



Energetic condition of recovered and resistant coral colonies five months after mass bleaching event.

(a) Total protein content normalised to host tissue surface area. (b) Total carbohydrate content normalised to host tissue surface area. The data point represents a single colony.



(c) Proportion of recovered and resistant colonies containing spermatozoa and oocytes. (d) Relative fecundity (fertility) in recovered and resistant colonies. Each data point represents one colony.

Fig 11.1

(a) Suggest which type of corals would demonstrate better promote reef recovery after disturbance. Justify your choice with evidence from Fig 11.1.

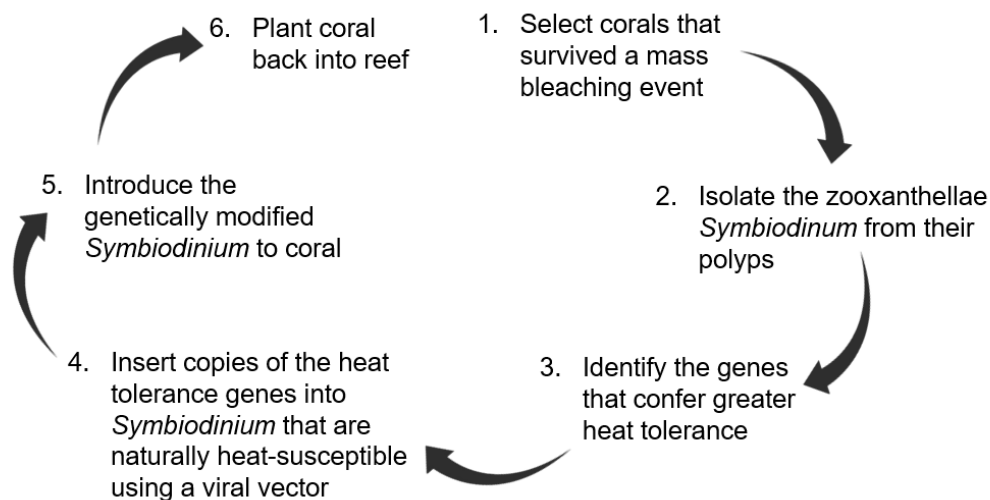
.....[3]

- 1 Resistant corals; [1]
- 2 ref. **higher energy reserves** which can be allocated for **coral reproduction / production of gametes**, to **repopulate coral reef**;
- 3 [Quote data for any one] i.e. higher total protein content (range of 250 – 650 ug/cm³) compared to recovered corals (200 – 550 ug/cm³)
/ higher carbohydrate content (range of 700 – 1700ug/cm³) compared to recovered corals (400 – 1600ug/cm³); **Accept ave/mean protein or carbohydrate content**
OR
- 4 [Quote data for any one] i.e. higher proportion of corals containing spermatocytes and oocytes (0.95) compared to recovered corals (0.35 and 0.30 respectively)
/ higher relative fecundity (3 ave oocytes per polyp) compared to recovered corals (0 ave oocytes per polyp);

Accept: Quoting of mean as comparison between resistant and recovered corals;

Synthetic biology offers the potential to isolate and cultivate the resistant strains of coral that can naturally withstand higher sea surface temperatures associated with climate change.

Fig. 11.2 shows the process of using synthetic biology in promoting community reef recovery.

**Fig. 11.2**

A national survey was conducted with 1,148 Australians, measuring psychological predictors of support for a synthetic biology conservation solution to coral loss.

Participants were moderately strongly supportive of technology development and were most keen to implement genetically engineered coral with between 50 and 70% of reef remaining intact.

The findings suggest that the general public are not averse to the development of a synthetic biology solution for restoring the reef, and they may be especially influenced by whether the synthetic biology solution is shown to be effective, particularly in comparison to other conservation solutions (such as rubble stabilization, manual removal of coral predators like crown-of-thorns starfish).

However, support for a synthetic biology intervention is conditional and many participants expressed concerns about possible long-term impacts on humans, animals, and the environment as a result of deploying engineered coral.

- (b) Discuss the ethical considerations that support and oppose the use of genetically engineered coral for reef restoration.

..... [2]

Support:

- 1 Restoration of reefs will restore the ecosystem services of the coral reef, such as coastal protection and nursery for marine species **despite a rising sea temperature**.
- 2 The synthetic biology solution will complement the existing conservation solutions to **hasten** the process of reef restoration. (Idea that rubble stabilization and manual removal of crown-of-thorn starfish will only reduce the death of existing coral, but it will take them time to recover in number, as opposed to deploying the genetically engineered coral to rebuild the reef; especially in the face of warming oceans).

Oppose:

- 3 Idea that there might be unforeseen adverse effect on the species that feed on the genetically engineered coral, and hence affects the whole food chain/web.
- 4 Idea that the genetically modified corals might out-compete the native corals and drive them to extinction. Leads to lower genetic diversity.
- 5 Idea that the selective breeding of heat-resistant coral can achieve the same result and is less risky.
- 6 The deployment of genetically modified coral might not be effective in restoring the reef's ecosystem services (idea that the reef might be rebuilt, but the marine species do not return to the reef)

[At least one point for each side of the argument]

Reject: Generalised statements like playing God, GE corals are unnatural, resulting in negative impacts on marine life etc. Suitable elaboration or explanations are needed.

[Total: 5]

End of Paper